LESSON 322 (1.0 GROUND)

Lesson Objective
During this ground lesson, the student will be introduced to loss of communication and equipment failure procedures.

Briefing
- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References
- Instrument Flying Handbook. (FAA-H-8083-15B)
- 14 CFR 91.185
- AIM 6-4

Lesson Content
Loss of Communication

- General procedures:

91.185 - Two-Way Radio Communications Failure

1. Troubleshoot
    - Return to last assigned frequency
    - Check volume of COM1/COM2 and headset
    - Unplug/replug headset, try different jacks
    - Reset COM1/COM2 circuit breaker
    - Test Squelch function
    - If you are familiar with the airport and have access to the control tower phone number, depending on if you have cellular network, call up the tower. Instructors, share MLB tower's phone number with your students: (321) 768-1524.
2. Squawk 7600
3. Determine if VMC (Visual Meteorological Conditions) or IMC (Instrument Meteorological Conditions:
    - If VMC, land as soon as practicable.
    - If IMC, determine a route and altitude to fly to the destination:

Route - in order:	Altitude - highest of:
Assigned	**M**inimum IFR Altitude
Vectored	**E**xpected
Expected	**A**ssigned
Filed	

        - If the clearance limit is a fix from which an approach begins, begin the approach at EFC (or ETA, if none exists).
        - If the clearance limit is not a fix from which an approach begins, leave clearance limit at EFC (or ETA, if none exists) and proceed to a fix from which an approach begins.
- During climb out:
    - Be aware of "dead vectors" - initial or temporary vectors ATC may assign with an intention to only fly momentarily.
        - "Fly runway heading"
        - "Fly heading ___ and contact departure"
        - "Fly heading ___ for traffic"
    - If the comms failure occurs after being given a "dead vector," use pilot discretion to revert down to Expected or Filed route.
        - Fly published SID/ODP.
        - If none, turn to the expected/filed route after reaching 400ft AGL.
        - Climb to at least OROCA to guarantee obstacle clearance for further turns.
- Holding:
    - If the aircraft reaches clearance limit before EFC/ETA, hold at the clearance limit until the EFC/ETA.
        - Holds & holding altitudes may be published for holds which are a part of the approach procedure (for example, SUROY).
        - Maintain the Minimum IFR Altitude throughout the hold. If holding over an IAF that is not part of the approach, the MSA (Minimum Safety Altitude) for that approach should be maintained.
- Approach:
    - Execute an approach to land as close as possible to the ETA.

Equipment Failure

- Attitude indicator
    - A failure of the **accelerometer** will result in a failure of the attitude indicator. Red Xs will be displayed on the PFD.
    - Reset the AHRS circuit breaker. The aircraft may need to be brought to a stop before final alignment can be made.
- Heading indicator
    - A failure of the **magnetometer** will result in a failure of the heading indicator. Red Xs will be displayed on the PFD.
    - Reset the AHRS circuit breaker. The aircraft may need to be brought to a stop before final alignment can be made.
- Turn coordinator
    - If the **electric turn coordinator** fails, a red flag will appear on the face of the instrument.
    - Reset the turn coordinator circuit breaker.

- Use of magnetic compass
    - The magnetic compass consists of two iron bars, attached to a face card, that sit atop a pivot, allowing the bars to align with the earth's magnetic field, displaying the aircraft's magnetic heading.
    - The iron bars are suspended in kerosene to prevent freezing.
    - Magnetic compass errors: **DVMONA**
        - **D**eviation: radio/equipment interference.
            - Corrected with the compass deviation card.
        - **V**ariation: difference between magnetic north and true north.
            - Corrected by applying an offset read from isogonic lines on VFR sectional.
        - **M**agnetic dip: tendency of the iron bar to dip down
        - **O**scillation: error caused by vibrations of the engine and movements of the aircraft.
        - **N**ortherly turning: tendency of the compass to lead/lag while turning to N/S.
            - Result of a counterweight attached to one end of the iron bars, used to reduce magnetic dip.
            - Corrected through **UNOS** - **U**ndershoot **N**orth, **O**vershoot **S**outh by approximately a heading value equal to the degrees of latitude while turning to north/south headings.
        - **A**cceleration: tendency of the compass to swing toward N/S while accelerating on E/W.
            - **ANDS** - **A**ccelerate **N**orth, **D**ecelerate **S**outh.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 323
- Assign Homework:
    - Review instrument approach procedures for ILS, VOR, and RNAV approaches.
    - Review partial panel basic attitude instrument procedures.

Completion Standards

This lesson shall be complete when through oral quizzing and discussion, the student displays a basic understanding of loss of communication and equipment failure procedures